

Medical Aid Charges: A Linear Regression Proof of Concept

Part 1 — Linear Regression Portfolio of Evidence

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1. Executive Summary

This report documents the development of a Linear Regression model that predicts medical charges from age, sex, BMI, number of children, smoker status and region, using a publicly available US insurance dataset as an initial proof of concept ahead of a South African pilot.

Key finding: smoker status is overwhelmingly the strongest driver of charges, followed by age and BMI. Sex and region were not statistically significant predictors and were removed. The final recommended model uses age, BMI, number of children and smoker status, and explains approximately 80% of the variance in charges on unseen data ($R\text{-squared} = 0.80$), with a typical prediction error of about \$4,200 (MAE).

2. Data Suitability and Exploratory Data Analysis

The dataset contains 1,338 records with no missing values. **charges** is a continuous, near-unique numeric variable, confirming this is a regression problem. Predictors are a mix of numeric (age, bmi, children) and categorical (sex, smoker, region) fields, all of which can be prepared for Linear Regression.

2.1 Target variable and outliers

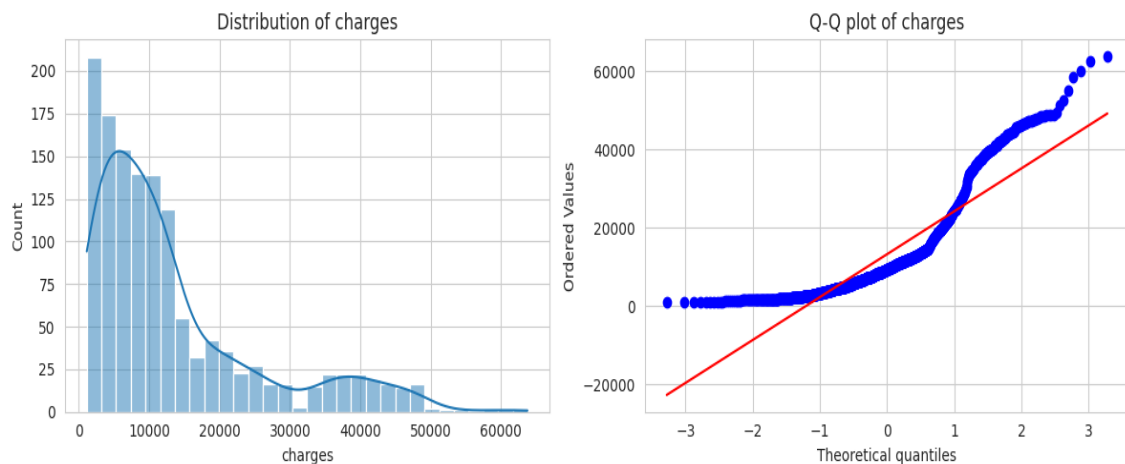


Figure 1: Distribution and Q-Q plot of charges — right-skewed with a long high-cost tail.

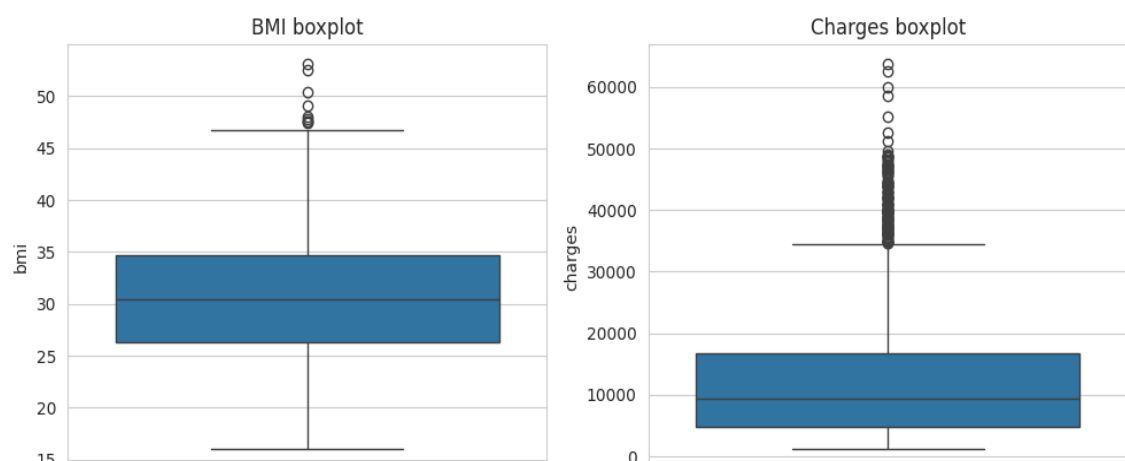


Figure 2: Boxplots of BMI and charges — a handful of high-BMI clients and many high-charge clients, both genuine rather than data errors.

One exact duplicate row was found and removed. No missing values were present. The high-BMI and high-charge values were judged to be genuine (clinically plausible obesity, and the expected cost profile of smokers) rather than data-entry errors, and were retained.

2.2 Relationships with charges

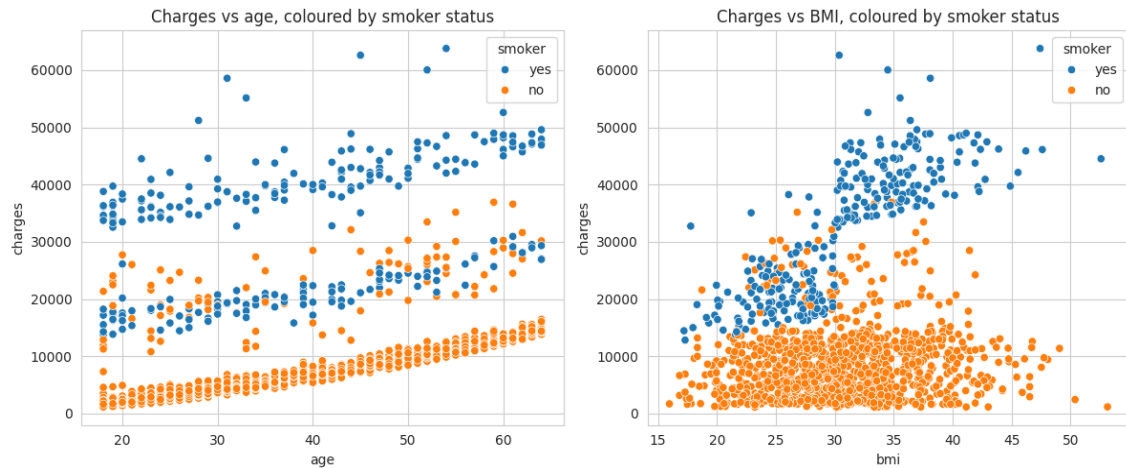


Figure 3: Charges vs age and BMI, coloured by smoker status.

Smokers form a visibly higher band of charges at every age and BMI level, and charges rise more steeply with BMI for smokers than non-smokers — an interaction later tested during retraining. Age shows a roughly linear, positive relationship with charges.

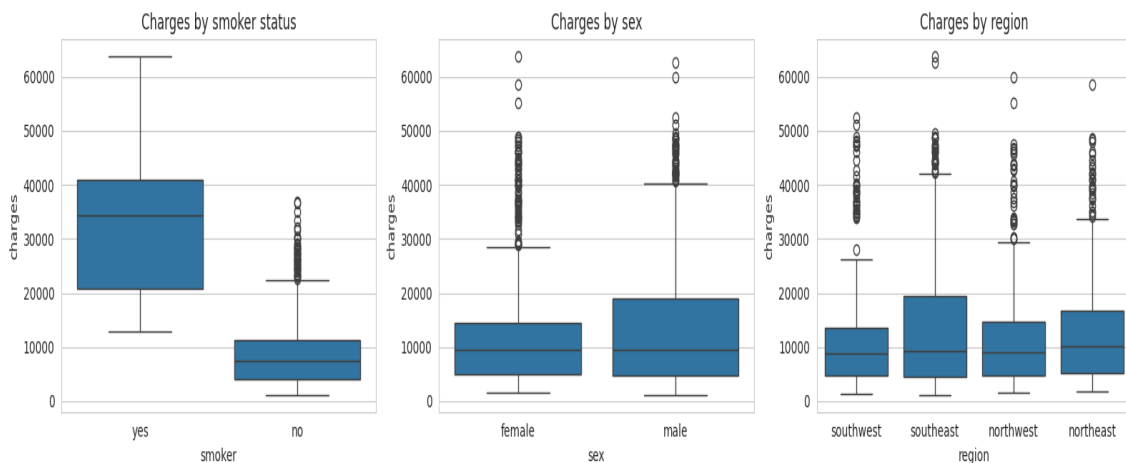


Figure 4: Charges by smoker status, sex and region.

Smoker status separates charges sharply. Sex and region show far less separation, foreshadowing their removal during feature selection.

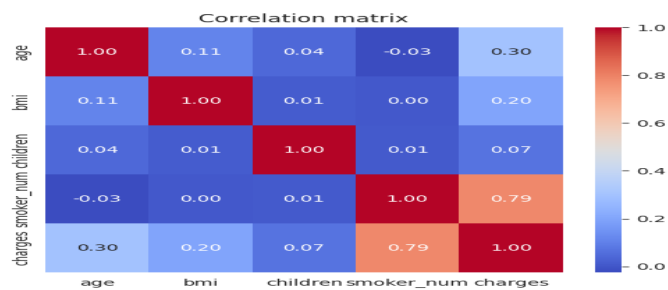


Figure 5: Correlation matrix of numeric features and charges.

Smoker status has the strongest correlation with charges (~0.79), followed by age (~0.30) and BMI (~0.20). Numeric predictors are not strongly correlated with each other, so multicollinearity between the raw numeric features is not a concern.

3. Feature Selection

Categorical variables were encoded (sex and smoker as 0/1; region one-hot encoded). A Variance Inflation Factor (VIF) check on the encoded features confirmed no problematic multicollinearity. **Backward elimination** was then applied: an Ordinary Least Squares model was fit repeatedly, removing the least significant feature (highest p-value above 0.05) each time, until only significant predictors remained.

- Removed (not significant): sex, region (all three dummy variables)
- Retained (significant at 5%): age, bmi, children, smoker

This matches the pattern seen in the EDA — sex and region carry little independent information about charges once age, BMI, children and smoker status are accounted for.

4. Model Training and Evaluation

An 80/20 train-test split (random seed 42) was used. Ordinary Least Squares Linear Regression has no hyperparameters to tune in the classic sense, so scikit-learn defaults were used for **Model 1**, trained on the four selected features against raw charges.

4.1 Evaluation metrics used

- **R-squared / Adjusted R-squared** — proportion of variance in charges explained; adjusted R-squared penalises unhelpful features.
- **RMSE** — error in dollars, penalising large mispricing errors more heavily (important for a scheme wanting to avoid badly under-pricing high-cost clients).
- **MAE** — a robust, easy-to-communicate typical dollar error.

4.2 Model 1 results

Metric	Value
R-squared	0.8046
Adjusted R-squared	0.8016
RMSE	\$5,992.88
MAE	\$4,198.59

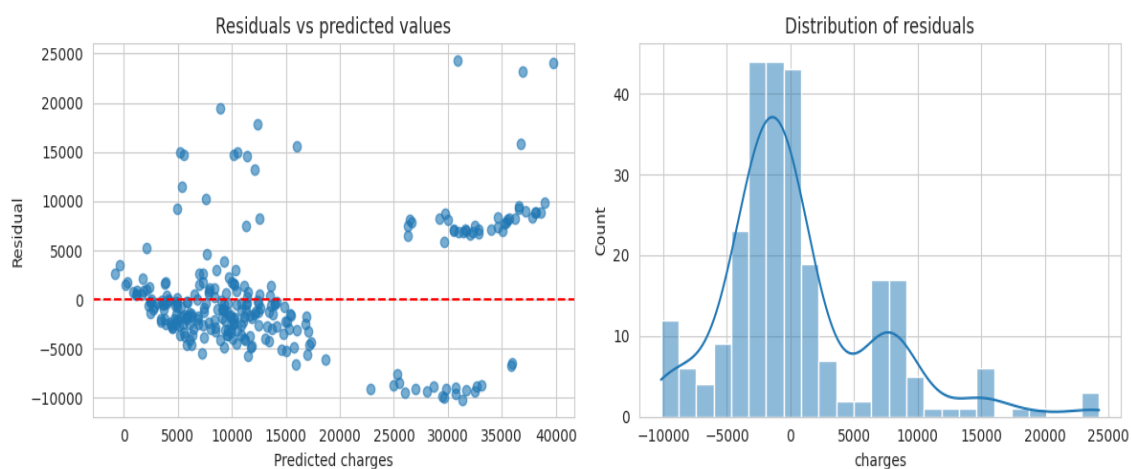


Figure 6: Model 1 residuals — a funnel/curved pattern shows the model under-predicts some high-cost clients and over-predicts some low-cost clients.

The residual pattern, together with the right-skew of charges seen earlier, indicates the Linear Regression assumption of homoscedastic, normally distributed residuals is violated. This motivated the retraining step

below.

5. Retraining and Model Comparison

Model 2 was trained on **log(charges)** with an added **bmi × smoker** interaction term, to address the skew and the steeper BMI effect observed for smokers. This produced a better fit in log space (R-squared = 0.83) with a far more symmetric residual pattern (Figure 7).

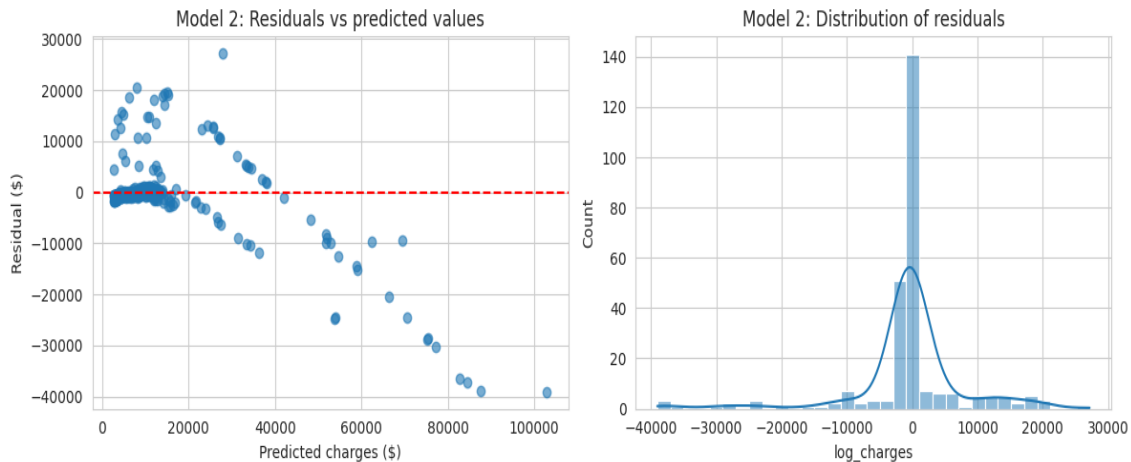


Figure 7: Model 2 residuals (dollar scale, naive back-transformation) — almost all residuals at higher predicted values are positive, visual evidence of the systematic under-prediction discussed below.

5.1 An important, honest finding: back-transformation bias

Simply exponentiating the log-space predictions to get back to dollars **worsened** performance (R-squared fell to 0.61, RMSE rose to \$8,465) — a classic retransformation bias, since exponentiating an average log-prediction under-states the true average for right-skewed data. A standard correction, **Duan's smearing estimator**, was applied (multiplying predictions by a correction factor of ~1.14). This removed the systematic bias, but — because non-smokers (cheap, tightly clustered charges) vastly outnumber smokers (expensive, widely spread charges) — the flat multiplicative correction over-corrected the low-cost majority, and RMSE and R-squared worsened further still (R-squared = 0.40, RMSE = \$10,513).

Metric	Model 1 (raw charges)	Model 2 (naive back-transform)	Model 2 (smearing-corrected)
R-squared	0.8046	0.6101	0.3985
Adjusted R-squared	0.8016	0.6026	0.3870
RMSE	\$5,992.88	\$8,464.67	\$10,513.11
MAE	\$4,198.59	\$4,144.86	\$5,305.62

Lesson: a theoretically "better" transformation can still under-perform in practice when the data is heteroscedastic across sub-groups, and models built on a transformed target must always be compared to alternatives on a common, correctly back-transformed scale rather than assumed to be an improvement.

6. Conclusion and Recommendation

- **Recommended model: Model 1** — plain Linear Regression on age, BMI, children and smoker status. It achieved the best R-squared (0.80) and RMSE (\$5,993) of the models tested, and is simple to explain to non-technical stakeholders.
- Smoker status is the dominant driver of medical charges, followed by age and BMI; sex and region were not significant and were dropped.

- The log-transform experiment (Model 2) is retained in the accompanying notebook as a documented, honest exploration that did not outperform the simpler model on this dataset — an important part of the analytical process, not a wasted step.
- This US dataset is a proof of concept only. Before deployment, the scheme should retrain the model on South African client and claims data, and consider non-linear models (e.g. tree-based regressors) to better capture the highest-cost clients.